

Nepal Import Review Draft - Grid-Connected PV Inverter

For SunBridge Trading, to share with the Nepal import agent. Checked against NEPQA-2025 §1.4. Draft for review.

1. Important: the two documents describe different products

Verdict: DIFFERENT_PRODUCT

Records do not share any model number and/or differ in phase. Their specifications are not comparable.

Field SUN-3K-G06P3-EU-AM2 (three-phase) CE-1P3001G-230-EU (single-phase)

Document type Certificate of Conformity Test Report

Phase three single

DC voltage class 1100V 60V

These two documents share a factory but are different product lines. They cannot be treated as one filing. Each is scored separately against NEPQA below.

Extracted facts, for reference - these are different products, not a conflict comparison:

Field	Value (A)	Value (B)	Sources
ac_output_voltage	230/400V	230V	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
certificate_or_report_number	PCS-24-1022	GZES230100125901	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
ip_rating	IP65	IP67	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
issue_date	2024-03-26	2023-02-01	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
manufacturer_address	No. 26 South YongJiang Road, Daqi, Beilun, NingBo, China	No. 1828 Fuqing South RD. Panhuo ST. Yinzhou District Ningbo Zhejiang 315000 China	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
manufacturer_name	NingBo Deye Inverter Technology Co., Ltd.	Zhejiang CHISAGE New Energy Technology Co., Ltd	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
output_frequency	50/60 Hz	50Hz	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
power_factor	0.8 leading~0.8 lagging	?0.99	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
rated_power	3kW, 4kW, 5kW, 6kW, 7kW, 8kW, 9kW, 10kW, 12kW, 15kW	300W to 2000W	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
standards	IEC 62116:2014 and IEC 61727:2004	IEC/EN 62109-1:2010	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf
test_lab	SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch	SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch	188_1115.pdf / DSS_GZES230100125901_combined-1.pdf

NEPQA §1.4 document scorecard - SUN-3K-G06P3-EU-AM2 (three-phase)

Models: SUN-3K-G06P3-EU-AM2, SUN-4K-G06P3-EU-AM2, SUN-5K-G06P3-EU-AM2, SUN-6K-G06P3-EU-AM2, SUN-7K-G06P3-EU-AM2, SUN-8K-G06P3-EU-AM2, SUN-9K-G06P3-EU-AM2, SUN-10K-G06P3-EU-AM2, SUN-12K-G06P3-EU-AM2, SUN-15K-G06P3-EU-AM2,

SUN-3K-G06P3-EU-AM2-P1,	SUN-4K-G06P3-EU-AM2-P1,	SUN-5K-G06P3-EU-AM2-P1,
SUN-6K-G06P3-EU-AM2-P1,	SUN-7K-G06P3-EU-AM2-P1,	SUN-8K-G06P3-EU-AM2-P1,
SUN-9K-G06P3-EU-AM2-P1,	SUN-10K-G06P3-EU-AM2-P1,	SUN-12K-G06P3-EU-AM2-P1,
SUN-15K-G06P3-EU-AM2-P1		

- MET - DOC-1: Test certificate to IEC 61727:2004 (utility interface characteristics), from an IEC/IEC-recognized body
- MET - DOC-2: Test certificate to IEC 62116:2014 (islanding prevention test procedure), from an IEC/IEC-recognized body
- MISSING - DOC-3: Test certificate to IEC 62891:2020 (MPPT efficiency of grid-connected inverters), from an IEC/IEC-recognized body
- MISSING - DOC-4: Test certificate to IEC 62109-1:2010 AND IEC 62109-2:2011 (safety: general + inverter-particular), from an IEC/IEC-recognized body
- ASK_FACTORY - DOC-5: Importer-manufacturer agreement, signed and stamped, stating the PV inverter warranty period
- ASK_FACTORY - DOC-6: Catalogue and technical datasheet of the PV inverter

NEPQA §1.4 technical scorecard - SUN-3K-G06P3-EU-AM2 (three-phase)

- MET - TECH-1: Rated AC output voltage: 3-phase 400V (L-L) $\pm 10\%$, or 1-phase 230V (L-N) $\pm 10\%$
- extracted: 230/400V (quote: "Nominal grid voltage (V) 3L/N/PE 230/400V")
- MET - TECH-2: Output frequency 50Hz $\pm 2.5\%$
- extracted: 50/60 Hz (quote: "Nominal grid frequency (Hz) 50/60")
- NOT_STATED - TECH-3: MPPT input efficiency $\geq 95\%$
- NOT_STATED - TECH-4: Inverter efficiency $\geq 95\%$ ($\leq 5\text{kVA}$) / $\geq 97\%$ ($> 5\text{kVA}$), transformerless
- NOT_STATED - TECH-5: Euro efficiency $\geq 94\%$ ($\leq 5\text{kVA}$) / $\geq 96\%$ ($> 5\text{kVA}$), transformerless; efficiency-vs-output curve provided
- NOT_STATED - TECH-6: Inverter efficiency $\geq 90\%$ for transformer topology
- NOT_STATED - TECH-7: No-load loss $< 0.5\%$ of rated power (transformerless)
- NOT_STATED - TECH-8: No-load loss $< 1.5\%$ of rated power (transformer topology)
- NOT_STATED - TECH-9: Total harmonic distortion (THD) $< 5\%$ at full load
- MET - TECH-10: Power factor > 0.99 at nominal power; adjustable 0.8 leading to 0.8 lagging
- extracted: 0.8 leading~0.8 lagging (quote: "Output power factor 0.8 leading~0.8 lagging")
- MET - TECH-11: Ingress protection $\geq \text{IP65}$ per IEC 60529
- extracted: IP65 (quote: "Protection degree IP65")
- NOT_STATED - TECH-12: Built-in meter and data logger with external user interface
- NOT_STATED - TECH-13: Internal protection: DC reverse polarity, grid fault, lightning on feeder
- NOT_STATED - TECH-14: Fully automatic operation (wake-up, synchronization, shutdown)
- NOT_STATED - TECH-15: Cooling: fan or heat sink to avoid excessive heating
- NOT_STATED - TECH-16: PV inverter warranty ≥ 5 years

Labeling - SUN-3K-G06P3-EU-AM2 (three-phase)

No label artwork or photo provided - pending from factory. What we could pre-fill from the extracted facts:

- LBL-1 (Name of the manufacturer): NingBo Deye Inverter Technology Co., Ltd.

- LBL-2 (Brand, Model and Type): SUN-3K-G06P3-EU-AM2, SUN-4K-G06P3-EU-AM2, SUN-5K-G06P3-EU-AM2, SUN-6K-G06P3-EU-AM2, SUN-7K-G06P3-EU-AM2, SUN-8K-G06P3-EU-AM2, SUN-9K-G06P3-EU-AM2, SUN-10K-G06P3-EU-AM2, SUN-12K-G06P3-EU-AM2, SUN-15K-G06P3-EU-AM2, SUN-3K-G06P3-EU-AM2-P1, SUN-4K-G06P3-EU-AM2-P1, SUN-5K-G06P3-EU-AM2-P1, SUN-6K-G06P3-EU-AM2-P1, SUN-7K-G06P3-EU-AM2-P1, SUN-8K-G06P3-EU-AM2-P1, SUN-9K-G06P3-EU-AM2-P1, SUN-10K-G06P3-EU-AM2-P1, SUN-12K-G06P3-EU-AM2-P1, SUN-15K-G06P3-EU-AM2-P1
- LBL-3 (Rated Power in Watt or VA): 3kW, 4kW, 5kW, 6kW, 7kW, 8kW, 9kW, 10kW, 12kW, 15kW
- LBL-4 (Input and output voltage (V) and frequency (Hz)): 230/400V, 50/60 Hz
- LBL-5 (Maximum Input Voltage): 1100V
- LBL-6 (MPPT Voltage Range): unknown - pending from factory
- LBL-7 (Serial Number): unknown - pending from factory

Manufacturer and test information - SUN-3K-G06P3-EU-AM2 (three-phase)

- Manufacturer: NingBo Deye Inverter Technology Co., Ltd.
- Manufacturer address: No. 26 South YongJiang Road, Daqi, Beilun, NingBo, China
- Certificate / report number: PCS-24-1022
- Issue date: 2024-03-26
- Test lab / issuing body: SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch

NEPQA §1.4 document scorecard - CE-1P3001G-230-EU (single-phase)

Models: CE-1P3001G-230-EU, CE-1P5001G-230-EU, CE-1P6001G-230-EU, CE-1P8001G-230-EU, CE-1P10001G-230-EU, CE-1P13001G-230-EU, CE-1P16001G-230-EU, CE-1P18001G-230-EU, CE-1P20001G-230-EU(1)

- MISSING - DOC-1: Test certificate to IEC 61727:2004 (utility interface characteristics), from an IECCEE/IECRE-listed body
- MISSING - DOC-2: Test certificate to IEC 62116:2014 (islanding prevention test procedure), from an IECCEE/IECRE-listed body
- MISSING - DOC-3: Test certificate to IEC 62891:2020 (MPPT efficiency of grid-connected inverters), from an IECCEE/IECRE-listed body
- MISSING - DOC-4: Test certificate to IEC 62109-1:2010 AND IEC 62109-2:2011 (safety: general + inverter-particular), from an IECCEE/IECRE-listed body
- ASK_FACTORY - DOC-5: Importer-manufacturer agreement, signed and stamped, stating the PV inverter warranty period
- ASK_FACTORY - DOC-6: Catalogue and technical datasheet of the PV inverter

NEPQA §1.4 technical scorecard - CE-1P3001G-230-EU (single-phase)

- MET - TECH-1: Rated AC output voltage: 3-phase 400V (L-L) $\pm 10\%$, or 1-phase 230V (L-N) $\pm 10\%$
- extracted: 230V (quote: "Rated grid voltage 230V")
- MET - TECH-2: Output frequency 50Hz $\pm 2.5\%$
- extracted: 50Hz (quote: "Rated grid frequency 50Hz")
- NOT_STATED - TECH-3: MPPT input efficiency ? 95%

- NOT_STATED - TECH-4: Inverter efficiency ? 95% (?5kVA) / ? 97% (>5kVA), transformerless
- NOT_STATED - TECH-5: Euro efficiency ? 94% (?5kVA) / ? 96% (>5kVA), transformerless; efficiency-vs-output curve provided
- NOT_STATED - TECH-6: Inverter efficiency ? 90% for transformer topology
- NOT_STATED - TECH-7: No-load loss < 0.5% of rated power (transformerless)
- NOT_STATED - TECH-8: No-load loss < 1.5% of rated power (transformer topology)
- NOT_STATED - TECH-9: Total harmonic distortion (THD) < 5% at full load
- MET - TECH-10: Power factor > 0.99 at nominal power; adjustable 0.8 leading to 0.8 lagging
- extracted: ?0.99 (quote: "Power factor ?0.99")
- MET - TECH-11: Ingress protection ? IP65 per IEC 60529
- extracted: IP67 (quote: "IP protection class.....: IP 67")
- NOT_STATED - TECH-12: Built-in meter and data logger with external user interface
- NOT_STATED - TECH-13: Internal protection: DC reverse polarity, grid fault, lightning on feeder
- NOT_STATED - TECH-14: Fully automatic operation (wake-up, synchronization, shutdown)
- NOT_STATED - TECH-15: Cooling: fan or heat sink to avoid excessive heating
- NOT_STATED - TECH-16: PV inverter warranty ? 5 years

Labeling - CE-1P3001G-230-EU (single-phase)

No label artwork or photo provided - pending from factory. What we could pre-fill from the extracted facts:

- LBL-1 (Name of the manufacturer): Zhejiang CHISAGE New Energy Technology Co., Ltd
- LBL-2 (Brand, Model and Type): CE-1P3001G-230-EU, CE-1P5001G-230-EU, CE-1P6001G-230-EU, CE-1P8001G-230-EU, CE-1P10001G-230-EU, CE-1P13001G-230-EU, CE-1P16001G-230-EU, CE-1P18001G-230-EU, CE-1P20001G-230-EU(1)
- LBL-3 (Rated Power in Watt or VA): 300W to 2000W
- LBL-4 (Input and output voltage (V) and frequency (Hz)): 230V, 50Hz
- LBL-5 (Maximum Input Voltage): 60V
- LBL-6 (MPPT Voltage Range): unknown - pending from factory
- LBL-7 (Serial Number): unknown - pending from factory

Manufacturer and test information - CE-1P3001G-230-EU (single-phase)

- Manufacturer: Zhejiang CHISAGE New Energy Technology Co., Ltd
- Manufacturer address: No. 1828 Fuqing South RD. Panhuo ST. Yinzhou District Ningbo Zhejiang 315000 China
- Certificate / report number: GZES230100125901
- Issue date: 2023-02-01
- Test lab / issuing body: SGS-CSTC Standards Technical Services Co., Ltd. Guangzhou Branch

To clarify with the factory

- Catalogue and technical datasheet of the PV inverter
- Importer-manufacturer agreement, signed and stamped, stating the PV inverter warranty period
- Test certificate to IEC 61727:2004 (utility interface characteristics), from an IECEE/IECRE-listed body
- Test certificate to IEC 62109-1:2010 AND IEC 62109-2:2011 (safety: general + inverter-particular), from an

IECEE/IECRE-listed body

- Test certificate to IEC 62116:2014 (islanding prevention test procedure), from an IECEE/IECRE-listed body
- Test certificate to IEC 62891:2020 (MPPT efficiency of grid-connected inverters), from an IECEE/IECRE-listed body

Statuses - documents: MET = standard found; MISSING = required standard not present; ASK_FACTORY = cannot be shown by this document type. Technical: MET/FAIL = checked against the extracted value; NOT_STATED = not in this document, await test reports.